Stress Detection based on TEO and MFCC speech features using Convolutional Neural Networks (CNN)

Muhammad Syafiq Nordin
Department of Electrical Engineering,
Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia
msyaf0911@gmail.com

*Ani Liza Asnawi
Department of Electrical Engineering,
Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia
*aniliza@iium.edu.my

Nur Aishah Zainal
Department of Electrical Engineering,
Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia
aishahzainal.school@gmail.com

Rashidah Funke Olanrewaju
Department of Electrical Engineering, Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia

Ahmad Zamani Jusoh
Department of Electrical Engineering, Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia

Siti Noorjannah Ibrahim
Department of Electrical Engineering, Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia

Nor Fadhillah Mohamed Azmin
Department of Chemical Engineering and Sustainability,
Kuliyyah of Engineering,
International Islamic University Malaysia
Kuala Lumpur, Malaysia

2022 IEEE International Conference on Computing (ICOCO) | 978-1-6654-8996-6/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICOCO56118.2022.10031771

***Abstract*— The effect of stress on mental and physical health is very concerning making it a fascinating and socially valuable field of study nowadays. Although a number of stress markers have been deployed, there are still issues involved with using these kinds of approaches. By developing a speech-based stress detection system, it could solve the problems faced by other currently available methods of detecting stress since it is a non-invasive and contactless approach. In this work, a fusion of Teager Energy Operator (TEO) and Mel Frequency Cepstral Coefficients (MFCC) namely Teager-MFCC (T-MFCC) are proposed as the speech features to be extracted from speech signals in recognizing stressed emotions. Since stressed emotions affect the nonlinear components of speech, TEO is applied to reflect the instantaneous energy of the components. Convolutional Neural Network (CNN) classifier is used with the proposed T- MFCC features on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) corpus. The proposed method (T-MFCC) had shown a better performance with classification accuracies of 95.83% and 95.37% for male and female speakers respectively compared to the MFCC feature extraction technique which achieves 84.26% (male) and 93.98% (female) classification accuracies.**

Keywords*— stress detection, Teager Energy Operator (TEO); Mel Frequency Cepstral Coefficient (MFCC); Teager-MFCC (T-MFCC); Convolutional Neural Network; speech features.

I. Introduction

Stress has undoubtedly become one of the main reasons that lead to major health issues, physically or mentally such as cardiovascular disease, depression, and sleep disorders [1] [2]. The World Health Organization (WHO) has recognized stress as the "Health Epidemic of the 21st Century" [3]. Since stress could bring countless negative consequences to humans, it makes stress a very concerning topic of investigation and a socially valuable field of study. Previous researchers have put a lot of work into producing reliable methods including blood samples, saliva samples, heart rate measurements, skin response, brain activity, and many more [4] [5] [6] [7]. These available methods had proven many times to be useful and effective in detecting stress and preventing its detrimental consequences.

However, some of these methods such as taking samples of blood, urine, and saliva are considered invasive and require a high level of expertise to perform. Such methods are usually done by professionals and well-trained practitioners in a lab and require a very lengthy analysis procedure [4]. Furthermore, conducting and measuring brain activity and skin response to detect stress requires very pricey equipment which usually can only be obtained at clinics and hospitals [8] [9]. Thus, access to such equipment is limited and the physical appearance of the patients at clinics or hospitals is crucial for the method to be implemented.

The speech signal produced by humans has generated much useful information. Each and every voice signal has a wealth of data about the speaker. Along with the linguistic information relevant to the message the speaker wants to convey, this data also includes paralinguistic information about the speaker's age, gender, and current emotional state. Due to the complexity of speech production and the significance of the physiological and cognitive systems that are involved in our (human) health and wellness, such as the brain and the respiratory system, small changes in a speaker's physical and mental state can affect their ability to control their vocal apparatus, frequently at a subconscious level. Even though detecting stress through speech analysis is considered the least invasive, least obtrusive, and least expensive among other available approaches, this method produced poor results compared to bio-signal analysis [10]. The application of intelligent signal analysis together with the machine learning approaches to decouple these various speech features with the goal of robustly and properly recognising them is the subject of extensive and ongoing study [11]. With the help of machine learning, it is believed to be able to improve the performance of speech analysis approaches in speech-stress detection tasks.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on December 02,2025 at 06:25:21 UTC from IEEE Xplore. Restrictions apply.

In this paper, a combination of the Teager Energy Operator (TEO) and Mel Frequency Cepstral Coefficient (MFFC), namely T-MFCC features, is proposed to produce a reliable speech-stress detection task. The proposed features reflect the prospects of fluctuation in the energy of airflow characteristics within the vocal tract [12]. A CNN classification model is used to classify the stressed emotions namely anger, disgust, sadness, and fear with reference to neutral speech from the RAVDESS dataset [13]. The proposed method's outcome is then compared with the result when using the MFCC feature only.

II. METHODOLOGY

A. System Design

The proposed system in this work utilized a feature fusion of TEO and MFCC namely T-MFCC features for the model input as well as CNN as the classification technique. Stressed emotions which are anger, disgust, sadness, and fear will be classified with reference to neutral speech from the RAVDESS dataset. Fig. 1. depicts a flowchart that illustrates the design flow of the proposed work.

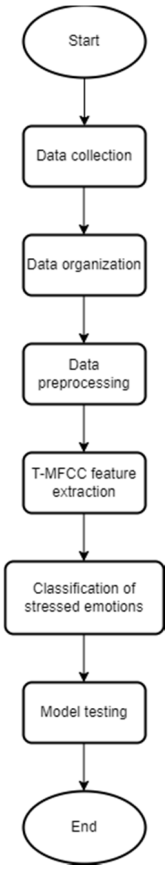

Fig. 1. Design flow of the proposed work

Throughout the implementation of this study, there are several software resources utilized including the Python programming language, the Jupyter notebook as the Integrated Development Environment (IDE), and the Python coding packages or libraries.

Python is a popular high-level programming language with an English-like syntax that makes the code easy to read and comprehend. The version of python that was used in this work is Python 3.9.7. Jupyter notebook was utilized in this work as an open-source IDE. Codes were run in small fragments to reduce the time for error detection and to get a quick output. Several Python libraries were used in completing this work namely Librosa, NumPy, Pandas, Tensorflow, and Scikit-learn.

B. Data Collection

The collection of speech data or audio samples is the most essential component of the stress identification system. In this work, speech samples were taken from the RAVDESS dataset. RAVDESS dataset is a well-known speech dataset that has been used especially in the speech recognition field. It consists of 24 professional actors (12 males, 12 females) vocalizing two lexically matched statements in a neutral North American accent. The database includes 8 emotions altogether namely neutral, calm, happy, sad, angry, fearful, disgust, and surprise. Even though the dataset comprises both speech and song audio files, only speech audio files were used in this study, with a total of 1440 speech audio files.

C. Data Organization

Data organization is crucially needed in this work to ensure that the raw speech samples from the RAVDESS database could be readable and easy to manage in this system. During data organization in this work, Pandas library played a vital role in creating a data frame for the raw speech samples. The raw speech samples were categorized in the data frame based on their speaker's identification number, speaker's gender, intensity, statement, repetition, and emotion acted.

D. Data Preprocessing

Before extracting features from the audio samples, an essential step to make sure these raw data are transformed into a useful and efficient format was implemented. The method is called data preprocessing. Fig. 2. shows the stages involved in data pre-processing.

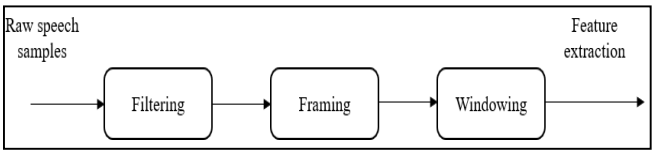

Fig. 2. Stages involved in data preprocessing

Filtering is a method of reducing noise in a speech signal that happens as a result of environmental disturbances or during the recording of a speech sample [12]. Since a speech signal is a non-stationary signal, analyzing it becomes more difficult. Thus, the framing method comes to ease the speech analysis process where the audio signal will be divided into an equal number of samples.

Since the audio was divided into several frames, there were some discontinuities at the edges of each frame which will create problems. To avoid this discontinuity, each frame was passed through a tapered window, in other words, the input signal was multiplied with a window function and the process is known as windowing.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on December 02,2025 at 06:25:21 UTC from IEEE Xplore. Restrictions apply.

E. T-MFCC feature extraction

Teager Energy Operator (TEO): There are linear and nonlinear components to speech production, and it has been determined that emotional speech, especially stressed speech, has significant differences in the nonlinear components from regular speech [14]. Teager has created an energy measurement to reflect the instantaneous energy of nonlinear components as a result of his speech study, and Kaiser initially described the energy operator's form as in the below equation [15]:

$$\psi[s(n)] = s2(n) - s(n+1)\, s(n-1) \tag{1}$$

Where $\psi s(n)$ is the Teager Energy Operator and $s(n)$ is the sampled speech signal.

Mel Frequency Cepstral Coefficients (MFCC): MFCC plays its role in mimicking the cochlea function using Mel filters. MFCC feature extraction technique includes all the preprocessing stages from transforming the frequency scale to the Mel scale as in the following equation [16]:

$$f_{mel} = 177\, ln\left(1 + \frac{fHz}{700}\right) \tag{2}$$

Where, f_{mel} is the frequency in Mel and f_{Hz} is the frequency in Hz. Next, MFCC has always been calculated using a series of triangular band-pass filters known as Mel Filter Bank were used to obtain the Mel spectrum by passing through the magnitude spectrum through it as shown in the equation (3) [16]:

$$H_m[k] = \begin{cases} 0 & k < f[m-1] \\ \frac{k - f[m-1]}{f[m] - f[m-1]} & f[m-1] < k \le f[m] \\ \frac{f[m+1] - k}{f[m+1] - f[m]} & f[m] < k \le f[m+1] \\ 0 & k > f[m+1] \end{cases} \tag{3}$$

Where, m = 0, 1, …, M-1. The following equation was later used to calculate the log-energy Mel spectrum [16]:

$$S[m] = ln\left[\sum_{k=0}^{N-1} |X[k]|^2 H_m[k]\right] \quad m = 0, 1, \cdots, M-1 \tag{4}$$

Where, $X[k]$ is the Discrete Fourier Transform (DFT) of an input speech signal $x[n]$. Finally, MFCC features were obtained by applying Discrete Cosine Transform (DCT) to the Mel spectrum (as depicted in the following equation), with the number of filters M ranging from 20 – 40, and the number of kept coefficients is 13 [16]:

$$\hat{x}[n] = \sum_{k=0}^{N-1} S[m] \cos\left[\left(m + \frac{1}{2}\right)\frac{\pi n}{M}\right], m = 0, \cdots, M-1 \tag{5}$$

TEO is used in conjunction with MFCC since stressed emotions are difficult to be identified with standard MFCC feature extraction [16]. In this work, speech signals have passed through the TEO block before they are preprocessed as in Fig. 3. This boosted the energy of the emotional content of the speech. Then, T-MFCC features of the speech samples were obtained by applying Discrete Fourier Transform (DFT), warping the frequencies on a Mel scale, and followed by applying the Discrete Cosine Transform (DCT) to the preprocessed signals.

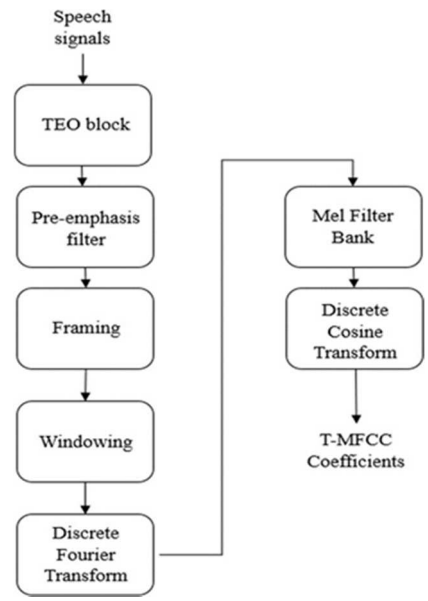

Fig. 3. T-MFCC feature extraction procedure

F. Classification of stressed emotions

After T-MFCC features were extracted from each speech sample, CNN was built with the help of the Keras and TensorFlow python libraries. A total of 100 epochs were set before the training started which means the training process will stop after the 100th epoch. Meanwhile, the batch size was set to 16 which means a total of 16 features from the training set are fed into the training model one at a time. The T-MFCC features of speech samples from the training set were fed into the training model. Fig. 4. depicts the flow of the implementation of the CNN training model in this work.

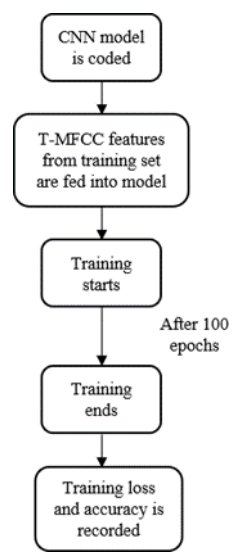

Fig. 4. Flow of the implementation of the CNN training model

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on December 02,2025 at 06:25:21 UTC from IEEE Xplore. Restrictions apply.

G. Model Testing

After the classification model was trained, the same model was utilized to evaluate how well its performance on unseen data or called the testing dataset. The data from the testing dataset were used to provide an unbiased evaluation of the final model fit on the training dataset. T-MFCC features from speech samples in the testing dataset were fed into the trained CNN model. The model classified the speech samples according to their emotions. Testing accuracy was recorded by comparing the actual class versus the predicted class of the speech samples in the testing dataset. For training the model, a total of 80% of the RAVDESS dataset's total speech samples were used, and the remaining 20% were used for testing.

III. Result and Analysis

A comprehensive analysis and discussion were performed based on the results generated from the methodology carried out in this work. The discussion was based on the performance of the proposed model in recognizing stressed emotions with reference to neutral speech using the proposed T-MFCC feature. The results demonstrated were divided according to the gender categories in the RAVDESS dataset. Finally, the proposed system was compared to the MFCC feature-based system.

For the male and female testing datasets, a total of 216 speech samples were used respectively. The testing datasets were randomized before the classification process took place, in order to make the classification result more reliable.

A. Stress Prediction with T-MFCC

To visualize and summarize the overall performance of the prediction results made by the model, a confusion matrix is constructed according to the gender categories. Fig. 5. shows the confusion matrix of stressed emotions classification on the male dataset while Fig. 6. is the confusion matrix of stressed emotions classification on the female dataset with T-MFCC.

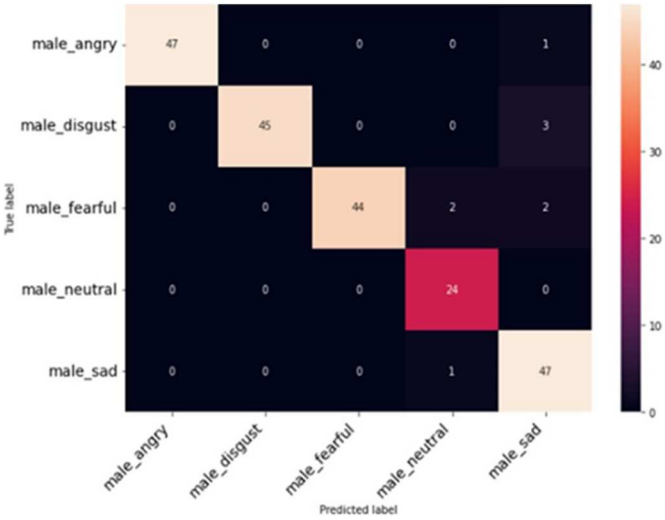

Fig. 5. Confusion matrix of stressed emotions classification on the male dataset using the T-MFCC feature

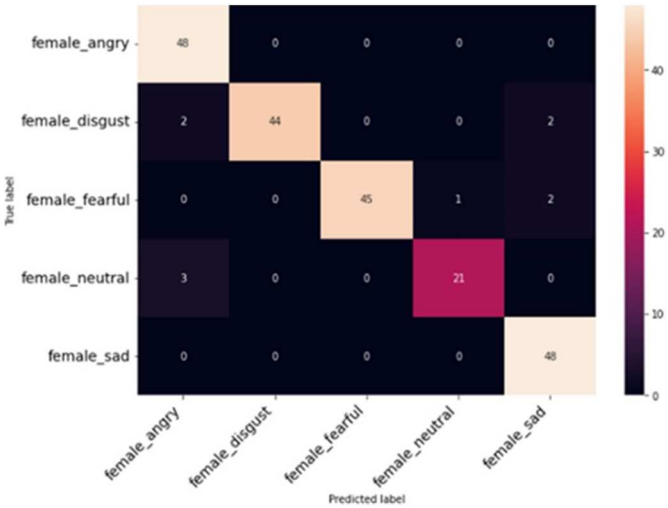

Fig. 6. Confusion matrix of stressed emotions classification on the female dataset using the T-MFCC feature

From the confusion matrices shown, it can be observed that the model has produced an almost perfect score for all the stressed and neutral emotions for both male and female datasets. The overall classification accuracy of the T-MFCC based stressed emotions recognition system for the male testing dataset is 95.83% while for the female dataset is 95.37%. The high percentage accuracies generated by the model for both male and female datasets indicate that T-MFCC features are effective in detecting stressed emotions regardless of the speaker's gender.

B. Comparison with the MFCC feature

In this work, MFCC features had also been extracted from the speech samples as the feature inputs for the classification model. This procedure was carried out in order to compare the system's performance when using the MFCC feature versus the proposed T-MFCC feature. Fig.7. depicts the confusion matrix of stressed emotions classification on the male dataset using the MFCC feature while Fig. 8. is for the female dataset.

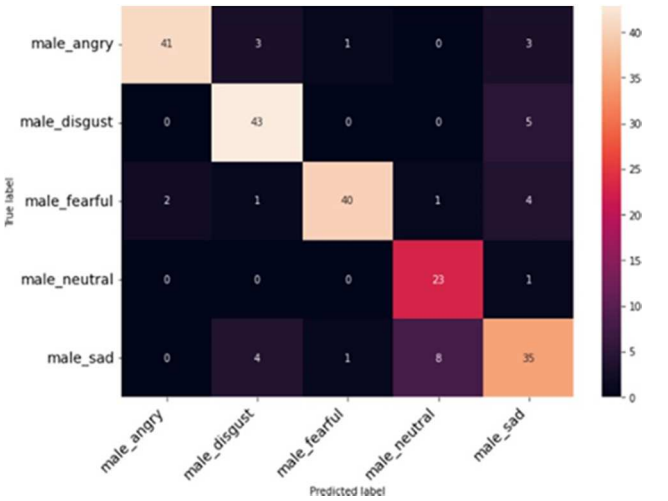

Fig. 7. Confusion matrix of stressed emotions classification on the male dataset using the MFCC feature only

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on December 02,2025 at 06:25:21 UTC from IEEE Xplore. Restrictions apply.

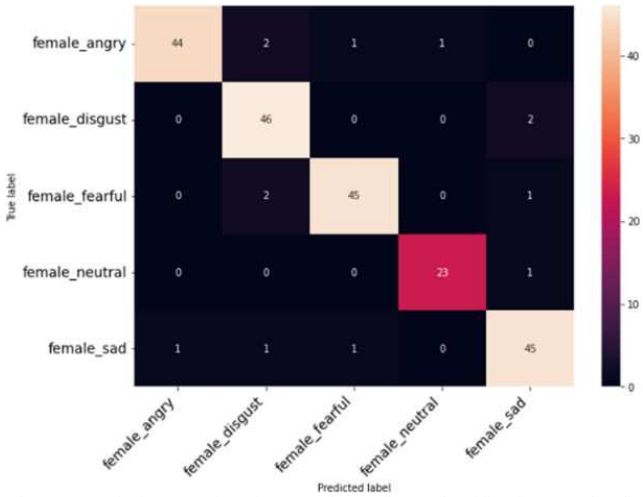

Fig. 8. Confusion matrix of stressed emotions classification on the female dataset using the MFCC feature only

Based on Fig. 7., it can be observed that the model has a lower prediction accuracy for males' sad emotions which is only 72.91% compared to other emotions which are above 80% of accuracy detection. This is corresponding to the reason justified in [17] that the MFCC features are not able to perfectly enhance the energies of the stressed emotions.

The overall classification accuracy of the stressed emotions recognition system using the MFCC features is 84.26% for the male dataset and 93.98% for the female dataset. This demonstrates that the MFCC feature performs better in classifying stressed emotions for females compared to males. Since the MFCC was built to enhance the energies of higher frequency components of speech signals [17], it performs better on females' speech signals because females speak with higher frequencies than males [18].

The following TABLE I. shows the comparison of the performance of the proposed T-MFCC-based system with the MFCC-based system.

TABLE I. COMPARISON OF PERFORMANCE OF THE PROPOSED T-MFCC-BASED SYSTEM WITH THE MFCC-BASED SYSTEM

Classification Accuracy (%)				
Emotion	**MFCC**		**T-MFCC**	
	Male	***Female***	***Male***	***Female***
Anger	85.42	91.67	97.92	100.00
Disgust	89.58	95.83	93.75	91.67
Fearful	83.33	93.75	91.67	93.75
Neutral	95.83	95.83	100.00	87.5
Sad	72.92	93.75	97.92	100.00
Overall Accuracy (based on Confusion Matrix)	**84.26**	**93.98**	**95.83**	**95.37**

Based on the table shown, the overall accuracy of the proposed T-MFCC feature-based system is higher than the MFCC feature-based system for both male and female speakers. It is clear that the accuracy of detecting stressed emotions is improved by introducing the TEO feature. Besides, the T-MFCC feature gives approximately the same classification performance regardless of the speaker's gender while the MFCC feature performs better on the female dataset compared to the male dataset. This is because MFCC was built to enhance the energies of higher frequency components of speech signals which makes it more effective in females' speech compared to males' speech. Furthermore, T-MFCC feature-based system gives high accuracies in detecting all stressed emotions in both male and female datasets whereas MFCC feature-based system has a low classification rate for the males' sad emotions. This is parallel with the findings in [17] which state that the TEO feature is used to enhance the energies of stressed emotions, which are not perfectly detected using MFCC basic feature extraction.

IV. CONCLUSION

In conclusion, speech-based stress detection can be the next most widely utilized method in helping people to detect stress and to restrain long-term stress consequences on affected people. The benefits of the speech feature-based stress detection system such as the non-invasive nature of the method, independence of assistance from technical experts as well as low-cost system equipment make the system preferable to any other existing methods.

The proposed T-MFCC feature was utilized as the feature input for the stressed emotions recognition system. The proposed system was able to identify all the stressed emotions namely anger, disgust, sadness, and fear with reference to neutral speech from the RAVDESS dataset via the CNN classification technique. An accuracy of 95.83% is observed using the proposed T-MFCC feature for the male dataset and 95.37% accuracy is achieved for the female dataset which is comparatively higher than that of MFCC alone with 84.26% and 93.98% respectively. The female dataset achieved better accuracy than the male due to the intensity of frequencies produced by female vocalizations being naturally higher than males. The study has proven that the energies stressed emotions could be enhanced by using the TEO feature and the combination of the two features which are TEO and MFCC (T-MFCC) does improve the recognitions significantly.

ACKNOWLEDGEMENT

This research was part of the work under the Ministry of Higher Education (MoHE) under the grant of Fundamental Research Grant Scheme (FRGS) with reference code FRGS/1/2021/TK0/UIAM/02/29.

REFERENCES

[1] R. L. C. Pan and J. K. J. Li, "A noninvasive parametric evaluation of stress effects on global cardiovascular function," *Cardiovasc. Eng.*, vol. 7, no. 2, pp. 74–80, 2007, doi: 10.1007/s10558-007-9028-6.

[2] A. R. Avila, S. R. Kshirsagar, A. Tiwari, D. Lafond, D. O'Shaughnessy, and T. H. Falk, "Speech-based stress classification based on modulation spectral features and convolutional neural

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on December 02,2025 at 06:25:21 UTC from IEEE Xplore. Restrictions apply.

networks," *Eur. Signal Process. Conf.*, vol. 2019-Septe, pp. 1–5, 2019, doi: 10.23919/EUSIPCO.2019.8903014.

[3] G. Fink, "Stress : Concepts , Cognition , Emotion , and Behavior : Handbook of Stress Stress : Concepts , Definition , and History George Fink Florey Institute of Neuroscience and Mental Health," no. March 2016, 2017.

[4] N. Sharma and T. Gedeon, "Objective measures, sensors and computational techniques for stress recognition and classification: A survey," *Comput. Methods Programs Biomed.*, vol. 108, no. 3, pp. 1287–1301, 2012, doi: 10.1016/j.cmpb.2012.07.003.

[5] A. Fernandes, R. Helawar, R. Lokesh, T. Tari and A. V. Shahapurkar, "Determination of stress using Blood Pressure and Galvanic Skin Response," 2014 International Conference on Communication and Network Technologies, 2014, pp. 165-168, doi: 10.1109/CNT.2014.7062747.

[6] P. Samal and R. Singla, "EEG Based Stress Level Detection During Gameplay," 2021 2nd Global Conference for Advancement in Technology (GCAT), 2021, pp. 1-4, doi: 10.1109/GCAT52182.2021.9587468.

[7] S. Betti et al., "Evaluation of an Integrated System of Wearable Physiological Sensors for Stress Monitoring in Working Environments by Using Biological Markers," in IEEE Transactions on Biomedical Engineering, vol. 65, no. 8, pp. 1748-1758, Aug. 2018, doi: 10.1109/TBME.2017.2764507.

[8] W. Liao, W. Zhang, Z. Zhu, and Q. Ji, "A Real-Time Human Stress Monitoring System Using Dynamic Bayesian Network," 2005.

[9] H. Peng, B. Hu, and F. Zheng, "A method of identifying chronic stress by EEG," 2012, doi: 10.1007/s00779-012- 0593-3.

[10] Vaikole, S., Mulajkar, S., More, A., Jayaswal, P., & Dhas Associate Professor, S. (2020). Stress Detection through Speech Analysis using Machine Learning. In *International Journal of Creative Research Thoughts* (Vol. 8). www.ijcrt.org

[11] N. Cummins, A. Baird, and B. W. Schuller, "Speech analysis for health: Current state-of-the-art and the increasing impact of deep learning," Methods, vol. 151, pp. 41–54, Dec. 2018, doi: 10.1016/j.ymeth.2018.07.007.

[12] X. Chen, H. Li, L. Ma, X. Liu, and J. Chen, "Teager-Mel and PLP fusion feature based speech emotion recognition," Proc. - 5th Int. Conf. Instrum. Meas. Comput. Commun. Control. IMCCC 2015, pp. 1109– 1114, 2016, doi: 10.1109/IMCCC.2015.239.

[13] S. R. Livingstone and F. A. Russo, "The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," 2018, doi: 10.5281/zenodo.1188976.

[14] H. Gao, S. Chen, P. An, and G. Su, "Emotion recognition of mandarin speech for different speech corpora based on nonlinear features," in 2012 IEEE 11th International Conference on Signal Processing, Oct. 2012, pp. 567–570. doi: 10.1109/ICoSP.2012.6491552.

[15] M. el Ayadi, M. S. Kamel, and F. Karray, "Survey on speech emotion recognition: Features, classification schemes, and databases," Pattern Recognit, vol. 44, no. 3, pp. 572–587, Mar. 2011, doi: 10.1016/j.patcog.2010.09.020.

[16] S. A. A. Qadri, T. S. Gunawan, M. Kartiwi, H. Mansor, and T. M. Wani, "Speech Emotion Recognition Using Feature Fusion of TEO and MFCC on Multilingual Databases," in Lecture Notes in Electrical Engineering, 2022, vol. 730, pp. 681–691. doi: 10.1007/978-981-33-4597-3_61.

[17] S. R. Bandela and T. K. Kumar, "Stressed speech emotion recognition using feature fusion of teager energy operator and MFCC," 8th Int. Conf. Comput. Commun. Netw. Technol. ICCCNT 2017, 2017, doi: 10.1109/ICCCNT.2017.8204149.

[18] M. Selvaraj, R. Bhuvana, and S. Padmaja, "Human speech emotion recognition," Int. J. Eng. Technol., vol. 8, no. 1, pp. 311–323, 2016

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on December 02,2025 at 06:25:21 UTC from IEEE Xplore. Restrictions apply.